

Lec: 13

LINEAR TRANSFORMATION

Gaussian Random Vectors

$$\bar{x} \sim N(\bar{\mu}, R)$$

$$E\{\bar{x}\} = \bar{\mu}$$

$$E\{(\bar{x} - \bar{\mu})(\bar{x} - \bar{\mu})^T\} = R.$$

$$\overset{m \times 1}{\bar{y}} = \underbrace{\overset{m \times n}{A} \overset{n \times 1}{\bar{x}} + \overset{m \times 1}{\bar{b}}}_{\text{Linear Transformation}}$$

Gaussian

Very interesting property:

Gaussian remains Gaussian under linear transformation

$$E\{\bar{y}\} = ?$$

$$E\{\bar{y}\} = E\{A\bar{x} + \bar{b}\} = A E\{\bar{x}\} + \bar{b}$$

Linear operator

$$\bar{\mu}_y = A \bar{\mu} + \bar{b}$$

mean of $\bar{y}$

$$E\{(\bar{y} - \bar{\mu}_y)(\bar{y} - \bar{\mu}_y)^T\}$$

covariance of $\bar{y}$.

$$= E \left\{ (A \bar{x} + \cancel{\bar{b}} - A \bar{\mu}_x - \cancel{\bar{b}}) (A \bar{x} + \cancel{\bar{b}} - A \bar{\mu}_x - \cancel{\bar{b}})^T \right\}$$

$$= E \left\{ (A(\bar{x} - \bar{\mu}_x)) (A(\bar{x} - \bar{\mu}_x))^T \right\}$$

$$= E \left\{ A(\bar{x} - \bar{\mu}_x) (\bar{x} - \bar{\mu}_x)^T A^T \right\}$$

$$= A E \left\{ (\bar{x} - \bar{\mu}_x) (\bar{x} - \bar{\mu}_x)^T \right\} A^T$$

$$= A R A^T$$


 Covariance matrix of $\bar{y}$.

$$\bar{y} \sim N(A \bar{\mu}_x + \bar{b}, A R A^T).$$

Special case : i.i.d Gaussian RVs

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \rightarrow \underbrace{\text{i.i.d.}}_{\substack{\text{independent} \\ \text{identically} \\ \text{distributed}}}$$

$$E \{ \bar{x} \} = 0$$

$$\bar{\mu}_x = 0.$$

$$E \{ \bar{x} \bar{x}^T \} = \sigma^2 \mathbf{I}$$

$$E\{x_i^2\} = \sigma^2 \quad \downarrow \text{Because i.i.d.}$$

Consider the linear transformation

$$y = \sum_{i=1}^n a_i x_i$$

$$y = \underbrace{a_1 x_1 + a_2 x_2 + \dots + a_n x_n}_{\text{weighted sum.}}$$

(Example:
Beamforming)

of i.i.d zero mean
Gaussian RVs.

$$= [a_1 \ a_2 \ \dots \ a_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= \bar{a}^T \bar{x}.$$

$$y = \bar{a}^T \bar{x}.$$

$$E\{\bar{a}^T \bar{x}\} = \bar{a}^T E\{\bar{x}\} = 0.$$

$$\begin{aligned} E\{y^2\} &= E\{(\bar{a}^T \bar{x})^2\} \\ &= E\{\bar{a}^T \bar{x} \bar{x}^T \bar{a}\} \\ &= \bar{a}^T \underbrace{E\{\bar{x} \bar{x}^T\}}_{\sigma^2 \mathbf{I}} \bar{a} \end{aligned}$$

$$\begin{aligned}
 E\{y^2\} &= \bar{a}^T \sigma^2 \mathbf{I} \bar{a} \\
 &= \sigma^2 \bar{a}^T \bar{a} \\
 &= \sigma^2 \|\bar{a}\|^2 \\
 &= \sigma^2 \sum_{i=1}^N a_i^2
 \end{aligned}$$

$$y \sim N(0, \sigma^2 \|\bar{a}\|^2)$$


 Beamforming vector

